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$$\text{list } 1 = \{ \quad \}$$

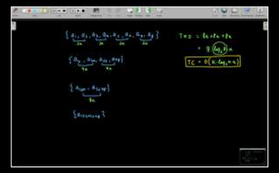
$$\text{list } 2 = \{ a_{01234567} \}$$

a_{67}, a_{45} $, a_{23}, a_{01}$

a_{0123}, a_{4567}

$, a_2, a_3$ $, a_4, a_5$

a_0, a_1



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Recursion → calls, callbacks

Backtracking

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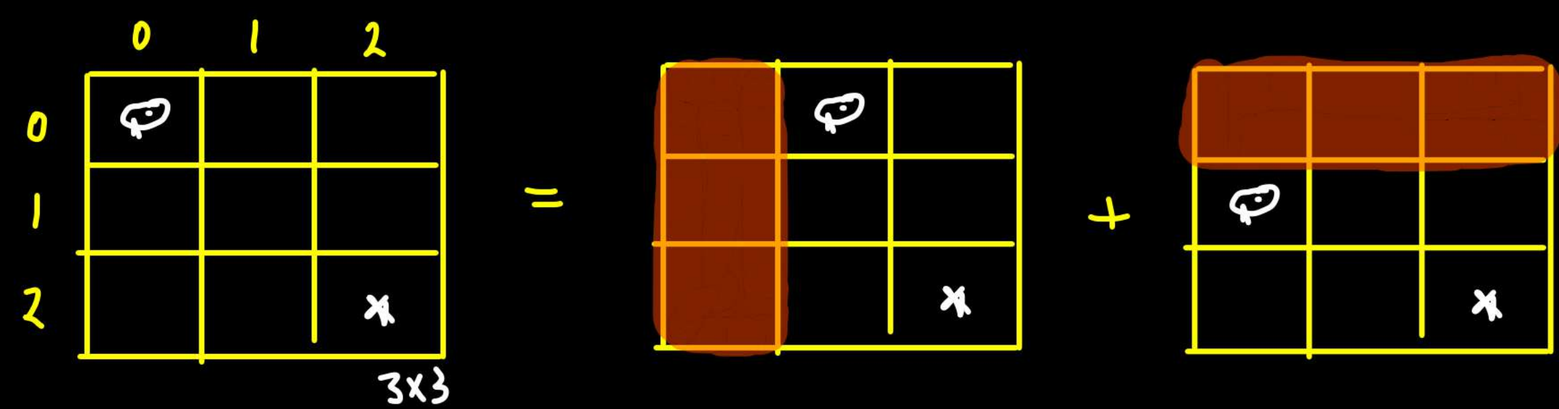
Ques: Rat in a Maze (4 directions)

no. of paths

down & right



$(0,0) \rightarrow (m-1, n-1)$

(r, c)

Andhadhun Recursion

right $\rightarrow c+1$

down $\rightarrow r+1$

```
path(r, c){
    if(r==m-1 && c==n-1) return 1;

    path(r+1,c)
    path(r, c+1);
```

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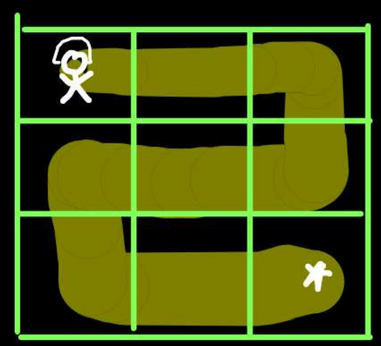
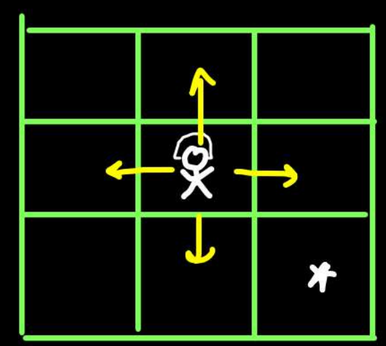
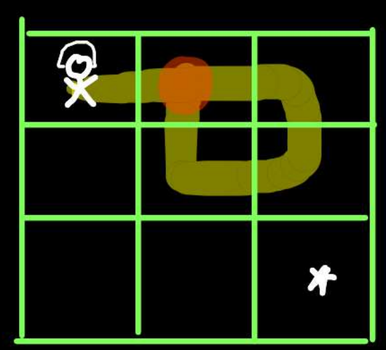
Down, Right, Up, Left (4 directions) → 4 calls



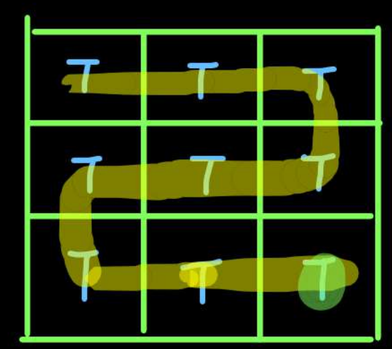
3x3



right = col+1
left = col-1
down = row+1
up = row-1



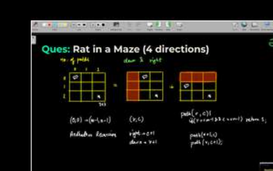
infinite calls
↓
solution ? → mark as visited



khaali cells are false

left
up
right
down

RRDLLDRR
~~RRDD~~



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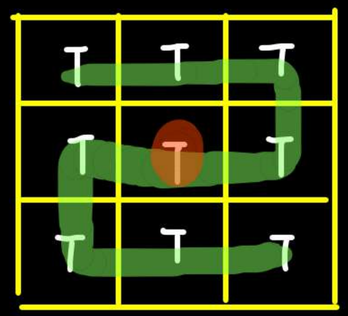
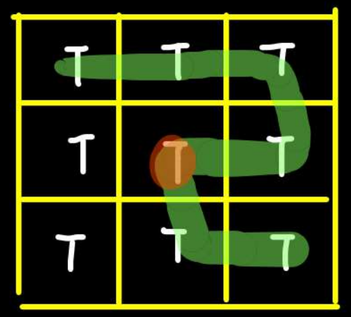
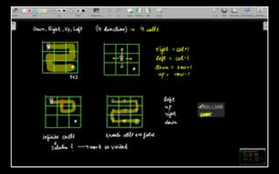


left
up
right
down



RRDLLDRR

Backtracking → calls ke baad wala kaam



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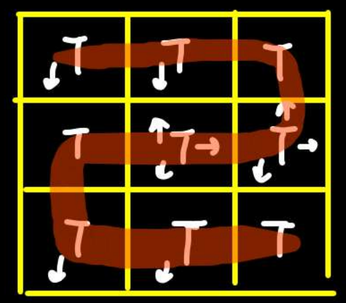
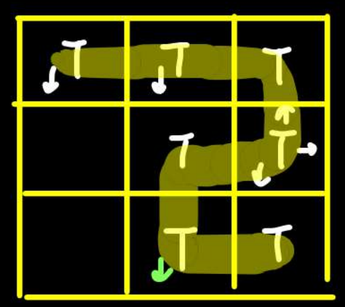


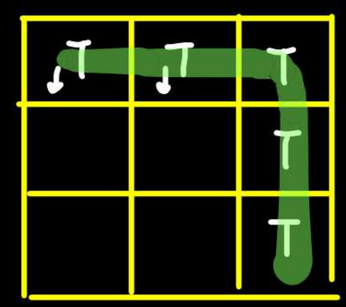
left
up
right
down

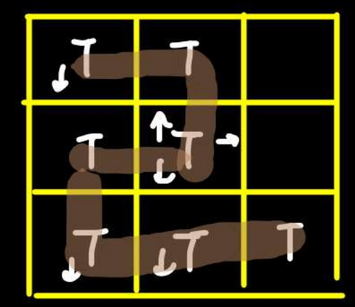
RRDLLDRR

RRDLDR

RRDD







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Ques: Rat in a Maze (Obstacles)

GFG Practice

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Ques: N Queens (Very Famous)

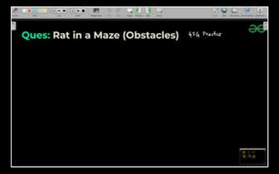
	Q		
			Q
Q			
		Q	

		Q	
Q			
			Q
	Q		

n = 4

	Q		
			Q
Q			
		Q	

Place 4 queens such that they can't kill each other

All 4 will be in diff. rows

```
if(canQueenBePlaced){
    mark Q
    move ahead (recursion)
    unmark Q // backtracking
}
```